NO MATH OR CODING REQUIRED

BIG-O

SAY IT
“BIG O”

SCAN & LEARN

HOW MUCH MORE WORK when there are MORE things?

4 ITEMS → 8 ITEMS — Twice as many!

ONE STEP
one small action you carry out

n
how many items to check

O(...)
how work grows

SAME 4 → 8 ITEMS. SIX WAYS THE AMOUNT OF WORK CAN GROW:

O(1) STAYS THE SAME

I know which drawer. I open it, however many drawers exist.

straight to known spot

4 items — 1 step → 8 items — 1 step

O(log n) GROWS VERY LITTLE

Search an ORDERED list. Discard half in each round.

4 › 2 › 1 — 2 rounds
8 › 4 › 2 › 1 — 3 rounds

4 items — 2 rounds → 8 items — 3 rounds

O(n) DOUBLES

Search drawers ONE BY ONE. If it is last, check all drawers.

1 2 3 4
1 2 3 4 5 6 7 8

4 items — 4 checks → 8 items — 8 checks

O(n log n) MORE THAN DOUBLES

Sort cards: split them into groups, then merge in order.

→ 2 rounds (of work)
→ 3 rounds (of work)

4 items — ~8 steps* → 8 items — ~24 steps*

O(n²) QUADRUPLES

Compare each item with EVERY item (including itself).

4 × 4 = 16 → 8 × 8 = 64

4 items — 16 checks → 8 items — 64 checks

O(2ⁿ) EXPLODES

For each item choose YES or NO. Try ALL possible combinations.

YES / NO — YES / NO

4 items — 16 ways → 8 items — 256 ways

REMEMBER JUST ONE QUESTION:

“If I double the items, HOW MUCH MORE WORK?”

same · a little · ×2 · over ×2 · ×4 · explodes

Big-O is not seconds or exact steps: it bounds how the workload grows.

* O(n log n) values show a simplified model, not measured steps.